STAFF REVIEW · H446 127 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. The instruction points to a location holding the effective address.

2. Sum 0, carry-out 1.

3. Linear/sequential search in the worst case.

4. RAM.

I Do — source worked steps

1. Medical image data — Predict the first step from the prompt.

2. Lossless encoding — Explain the intermediate result or decision.

3. Exact reconstruction — Lossless compression is required if exact original image data must be preserved. Lossy compression permanently discards information and could alter diagnostically important features.

We Do — guided CHECK

1. Limited bandwidth
Predict the first step from the prompt.
Choose acceptable bitrate
Explain the intermediate result or decision.
Consider visual quality
Not necessarily. Perceptually lossy encoding can reduce bitrate enough for smooth playback, but visible artefacts may appear. Suitability depends on quality, latency and the application.

You Do — source mark schemes

1. Lossless preserves exact data [2] with source code/archives [1]; lossy irreversibly discards detail [2] with streaming images/audio/video [1].

2. 12:3 reduces to 4:1 [4], or factor 4.

3. Source code lossless [2], photo possibly lossy [2], diagnostic scan lossless if exact data required [2], trade-off and qualified reasoning [2].

Exit Ticket — source answer

1. False. Lossy compression irreversibly discards some information.